

FRBNY DSGE Model Documentation

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1 General Structure

The FRBNY DSGE model is a medium scale, one-sector dynamic stochastic general equilibrium model which is based on the New Keynesian model with financial frictions used in Del Negro et al. (2015). The core of the model is based on the work of Smets and Wouters (2007) (henceforth SW) and Christiano et al. (2005): It builds on the neo-classical growth model by adding nominal wage and price rigidities, variable capital utilization, costs of adjusting investment, habit formation in consumption. The model also includes credit frictions as in the *financial accelerator* model developed by Bernanke et al. (1999b), where the actual implementation of the credit frictions follows closely Christiano et al. (2014), and a time-varying inflation target following Del Negro and Schorfheide (2012). In contrast to these papers, the model features both a deterministic and a stochastic trend in productivity. Finally, it accounts for forward guidance in monetary policy by including anticipated policy shocks as in Laseen and Svensson (2011).

The model economy is populated by eight classes of agents: 1) a continuum of households, who consume and supply differentiated labor; 2) competitive labor aggregators that combine labor supplied by individual households; 3) competitive final good-producing firms that aggregate the intermediate goods into a final product; 4) a continuum of monopolistically competitive intermediate good producing firms; 5) competitive capital producers that convert final goods into capital; 6) a continuum of entrepreneurs who purchase capital using both internal and borrowed funds and rent it to intermediate good producing firms; 7) a representative bank collecting deposits from the households and lending funds to the entrepreneurs; and finally 8) a government, composed of a monetary authority that sets short-term interest rates and a fiscal authority that sets public spending and collects taxes.

2 DSGE Model Specification

Since the derivation of the SW model is discussed in detail in Christiano et al. (2005), we will only present a summary of the log-linearized equilibrium conditions. We will first

reproduce the equilibrium conditions for the SW model and then discuss the two extensions that underly the DSGE model used for our analysis.

1.1 The SW Model

All variables in the following equations are expressed in log deviations from their non-stochastic steady state. In SW this is defined as $Z_t = e^{\gamma t + \frac{1}{1-\alpha} \tilde{z}_t}$, where γ is the steady state growth rate of the economy, and $\tilde{z}_t$ be the linearly detrended log productivity process which follows the autoregressive law of motion

$$\tilde{z}_t = \rho_z \tilde{z}_{t-1} + \sigma_z \varepsilon_{z,t}. \quad (1)$$

The growth rate of Z_t in deviations from γ , denoted by z_t , follows the process:

$$z_t = \ln(Z_t/Z_{t-1}) - \gamma = \frac{1}{1-\alpha}(\rho_z - 1)\tilde{z}_{t-1} + \frac{1}{1-\alpha}\sigma_z \varepsilon_{z,t}. \quad (2)$$

Steady state values are denoted by *-subscripts and steady state formulas are provided in the technical appendix of Del Negro and Schorfheide (2012), which is available online. The consumption Euler equation is given by:

$$\begin{aligned} c_t = & -\frac{(1 - he^{-\gamma})}{\sigma_c(1 + he^{-\gamma})} (R_t - \mathbb{E}_t[\pi_{t+1}] + b_t) + \frac{he^{-\gamma}}{(1 + he^{-\gamma})} (c_{t-1} - z_t) \\ & + \frac{1}{(1 + he^{-\gamma})} \mathbb{E}_t[c_{t+1} + z_{t+1}] + \frac{(\sigma_c - 1)}{\sigma_c(1 + he^{-\gamma})} \frac{w_* L_*}{c_*} (L_t - \mathbb{E}_t[L_{t+1}]), \end{aligned} \quad (3)$$

where c_t is consumption, L_t is labor supply, R_t is the nominal interest rate, and π_t is inflation. The exogenous process b_t drives a wedge between the intertemporal ratio of the marginal utility of consumption and the riskless real return $R_t - \mathbb{E}_t[\pi_{t+1}]$, and follows an AR(1) process with parameters ρ_b and σ_b . The parameters σ_c and h capture the degree of relative risk aversion and the degree of habit persistence in the utility function, respectively. The following condition expresses the relationship between the value of capital in terms of consumption q_t^k and the level of investment i_t measured in terms of consumption goods:

$$q_t^k = S'' e^{2\gamma} (1 + \bar{\beta}) \left(i_t - \frac{1}{1 + \bar{\beta}} (i_{t-1} - z_t) - \frac{\bar{\beta}}{1 + \bar{\beta}} \mathbb{E}_t[i_{t+1} + z_{t+1}] - \mu_t \right), \quad (4)$$

which is affected by both investment adjustment cost (S'' is the second derivative of the adjustment cost function) and by μ_t , an exogenous process called the “marginal efficiency of investment” that affects the rate of transformation between consumption and installed capital (see Greenwood et al. (1998)). The exogenous process μ_t follows an AR(1) process with parameters ρ_μ and σ_μ . The parameter $\bar{\beta} = \beta e^{(1-\sigma_c)\gamma}$ depends on the intertemporal discount rate in the utility function of the households β , the degree of relative risk aversion σ_c , and the steady-state growth rate γ .

The capital stock, $\bar{k}_t$, evolves as

$$\bar{k}_t = \left(1 - \frac{i_*}{\bar{k}_*}\right) (\bar{k}_{t-1} - z_t) + \frac{i_*}{\bar{k}_*} i_t + \frac{i_*}{\bar{k}_*} S'' e^{2\gamma} (1 + \bar{\beta}) \mu_t, \quad (5)$$

where $i_*/\bar{k}_*$ is the steady state ratio of investment to capital. The arbitrage condition between the return to capital and the riskless rate is:

$$\frac{r_*^k}{r_*^k + (1 - \delta)} \mathbb{E}_t[r_{t+1}^k] + \frac{1 - \delta}{r_*^k + (1 - \delta)} \mathbb{E}_t[q_{t+1}^k] - q_t^k = R_t + b_t - \mathbb{E}_t[\pi_{t+1}], \quad (6)$$

where r_t^k is the rental rate of capital, r_*^k its steady state value, and δ the depreciation rate. Given that capital is subject to variable capacity utilization u_t , the relationship between $\bar{k}_t$ and the amount of capital effectively rented out to firms k_t is

$$k_t = u_t - z_t + \bar{k}_{t-1}. \quad (7)$$

The optimality condition determining the rate of utilization is given by

$$\frac{1 - \psi}{\psi} r_t^k = u_t, \quad (8)$$

where ψ captures the utilization costs in terms of foregone consumption. Real marginal costs for firms are given by

$$mc_t = w_t + \alpha L_t - \alpha k_t, \quad (9)$$

where w_t is the real wage and α is the income share of capital (after paying mark-ups and fixed costs) in the production function. From the optimality conditions of goods producers

it follows that all firms have the same capital-labor ratio:

$$k_t = w_t - r_t^k + L_t. \quad (10)$$

The production function is:

$$y_t = \Phi_p (\alpha k_t + (1 - \alpha)L_t) + \mathcal{I}\{\rho_z < 1\}(\Phi_p - 1) \frac{1}{1 - \alpha} \tilde{z}_t, \quad (11)$$

if the log productivity is trend stationary. The last term $(\Phi_p - 1) \frac{1}{1 - \alpha} \tilde{z}_t$ drops out if technology has a stochastic trend, because in this case one has to assume that the fixed costs are proportional to the trend. Similarly, the resource constraint is:

$$y_t = g_t + \frac{c_*}{y_*} c_t + \frac{i_*}{y_*} i_t + \frac{r_*^k k_*}{y_*} u_t - \mathcal{I}\{\rho_z < 1\} \frac{1}{1 - \alpha} \tilde{z}_t, \quad (12)$$

where again the term $-\frac{1}{1 - \alpha} \tilde{z}_t$ disappears if technology follows a unit root process. Government spending g_t is assumed to follow the exogenous process:

$$g_t = \rho_g g_{t-1} + \sigma_g \varepsilon_{g,t} + \eta_{gz} \sigma_z \varepsilon_{z,t}.$$

Finally, the price and wage Phillips curves are, respectively:

$$\pi_t = \kappa mc_t + \frac{\iota_p}{1 + \iota_p \bar{\beta}} \pi_{t-1} + \frac{\bar{\beta}}{1 + \iota_p \bar{\beta}} \mathbb{E}_t[\pi_{t+1}] + \lambda_{f,t}, \quad (13)$$

and

$$\begin{aligned} w_t = \frac{(1 - \zeta_w \bar{\beta})(1 - \zeta_w)}{(1 + \bar{\beta})\zeta_w((\lambda_w - 1)\epsilon_w + 1)} (w_t^h - w_t) - \frac{1 + \iota_w \bar{\beta}}{1 + \bar{\beta}} \pi_t + \frac{1}{1 + \bar{\beta}} (w_{t-1} - z_t - \iota_w \pi_{t-1}) \\ + \frac{\bar{\beta}}{1 + \bar{\beta}} \mathbb{E}_t[w_{t+1} + z_{t+1} + \pi_{t+1}] + \lambda_{w,t}, \end{aligned} \quad (14)$$

where $\kappa = \frac{(1 - \zeta_p \bar{\beta})(1 - \zeta_p)}{(1 + \iota_p \bar{\beta})\zeta_p((\Phi_p - 1)\epsilon_p + 1)}$, the parameters ζ_p , ι_p , and ϵ_p are the Calvo parameter, the degree of indexation, and the curvature parameter in the Kimball aggregator for prices, and ζ_w , ι_w , and ϵ_w are the corresponding parameters for wages. w_t^h measures the household's

marginal rate of substitution between consumption and labor, and is given by:

$$w_t^h = \frac{1}{1 - h e^{-\gamma}} (c_t - h e^{-\gamma} c_{t-1} + h e^{-\gamma} z_t) + \nu_l L_t, \quad (15)$$

where ν_l characterizes the curvature of the disutility of labor (and would equal the inverse of the Frisch elasticity in absence of wage rigidities). The mark-ups $\lambda_{f,t}$ and $\lambda_{w,t}$ follow exogenous ARMA(1,1) processes

$$\lambda_{f,t} = \rho_{\lambda_f} \lambda_{f,t-1} + \sigma_{\lambda_f} \varepsilon_{\lambda_f,t} + \eta_{\lambda_f} \sigma_{\lambda_f} \varepsilon_{\lambda_f,t-1}, \text{ and}$$

$$\lambda_{w,t} = \rho_{\lambda_w} \lambda_{w,t-1} + \sigma_{\lambda_w} \varepsilon_{\lambda_w,t} + \eta_{\lambda_w} \sigma_{\lambda_w} \varepsilon_{\lambda_w,t-1},$$

respectively. Finally, the monetary authority follows a generalized feedback rule:

$$R_t = \rho_R R_{t-1} + (1 - \rho_R) \left(\psi_1 \pi_t + \psi_2 (y_t - y_t^f) \right) + \psi_3 \left((y_t - y_t^f) - (y_{t-1} - y_{t-1}^f) \right) + r_t^m, \quad (16)$$

where the flexible price/wage output y_t^f is obtained from solving the version of the model without nominal rigidities (that is, Equations (3) through (12) and (15)), and the residual r_t^m follows an AR(1) process with parameters ρ_{r^m} and σ_{r^m} .

1.2 Time-Varying Target Inflation and Long-Run Inflation Expectations

In order to capture the rise and fall of inflation and interest rates in the estimation sample, we replace the constant target inflation rate by a time-varying target inflation. While time-varying target rates have been frequently used for the specification of monetary policy rules in DSGE model (e.g., Erceg and Levin (2003) and Smets and Wouters (2003), among others), we follow the approach of Aruoba and Schorfheide (2008) and Del Negro and Eusepi (2011) and include data on long-run inflation expectations as an observable into the estimation of the DSGE model. At each point in time, the long-run inflation expectations essentially determine the level of the target inflation rate. To the extent that long-run inflation expectations at the forecast origin contain information about the central bank's objective function, e.g. the desire to stabilize inflation at 2%, this information is automatically included in the forecast.

More specifically, for the SW model the interest-rate feedback rule of the central bank (16) is modified as follows:

$$\begin{aligned} R_t = & \rho_R R_{t-1} + (1 - \rho_R) \left(\psi_1(\pi_t - \pi_t^*) + \psi_2(y_t - y_t^f) \right) \\ & + \psi_3 \left((y_t - y_t^f) - (y_{t-1} - y_{t-1}^f) \right) + r_t^m. \end{aligned} \quad (17)$$

The time-varying inflation target evolves according to:

$$\pi_t^* = \rho_{\pi^*} \pi_{t-1}^* + \sigma_{\pi^*} \epsilon_{\pi^*,t}, \quad (18)$$

where $0 < \rho_{\pi^*} < 1$ and $\epsilon_{\pi^*,t}$ is an iid shock. We model π_t^* as a stationary process, although our prior for ρ_{π^*} will force this process to be highly persistent. The assumption that the changes in the target inflation rate are exogenous is, to some extent, a short-cut. For instance, the learning models of Sargent (1999) or Primiceri (2006) imply that the rise in the target inflation rate in the 1970's and the subsequent drop is due to policy makers learning about the output-inflation trade-off and trying to set inflation optimally. We are abstracting from such a mechanism in our specification.

1.3 Financial Frictions

Building on the work of Bernanke et al. (1999a), Christiano et al. (2003), De Graeve (2008), and Christiano et al. (2014) we also add financial frictions to our DSGE model. We assume that banks collect deposits from households and lend to entrepreneurs who use these funds as well as their own wealth to acquire physical capital, which is rented to intermediate goods producers. Entrepreneurs are subject to idiosyncratic disturbances that affect their ability to manage capital. Their revenue may thus be too low to pay back the bank loans. Banks protect themselves against default risk by pooling all loans and charging a spread over the deposit rate. This spread may vary as a function of the entrepreneurs' leverage and their riskiness. Adding these frictions to the SW model amounts to replacing equation (6) with the following conditions:

$$E_t \left[\tilde{R}_{t+1}^k - R_t \right] = b_t + \zeta_{sp,b} (q_t^k + \bar{k}_t - n_t) + \tilde{\sigma}_{\omega,t} \quad (19)$$

and

$$\tilde{R}_t^k - \pi_t = \frac{r_*^k}{r_*^k + (1 - \delta)} r_t^k + \frac{(1 - \delta)}{r_*^k + (1 - \delta)} q_t^k - q_{t-1}^k, \quad (20)$$

where $\tilde{R}_t^k$ is the gross nominal return on capital for entrepreneurs, n_t is entrepreneurial equity, and $\tilde{\sigma}_{\omega,t}$ captures mean-preserving changes in the cross-sectional dispersion of ability across entrepreneurs (see Christiano et al. (2014)) and follows an AR(1) process with parameters ρ_{σ_ω} and σ_{σ_ω} . The second condition defines the return on capital, while the first one determines the spread between the expected return on capital and the riskless rate. Note that if $\zeta_{sp,b} = 0$ and the financial friction shocks $\tilde{\sigma}_{\omega,t}$ are zero, (19) and (20) coincide with (6). The following condition describes the evolution of entrepreneurial net worth:

$$\begin{aligned} n_t = & \zeta_{n,\tilde{R}^k} \left(\tilde{R}_t^k - \pi_t \right) - \zeta_{n,R} (R_{t-1} - \pi_t) + \zeta_{n,qK} (q_{t-1}^k + \bar{k}_{t-1}) + \zeta_{n,n} n_{t-1} \\ & - \frac{\zeta_{n,\sigma_\omega}}{\zeta_{sp,\sigma_\omega}} \tilde{\sigma}_{\omega,t-1}. \end{aligned} \quad (21)$$

1.4 Adding Long Run Changes in Productivity Growth

We add long run changes in productivity and now define Z_t as

$$Z_t = e^{\frac{1}{1-\alpha} \tilde{z}_t} Z_t^p e^{(\gamma + \frac{\alpha}{1-\alpha} \log \Upsilon) t}, \quad (22)$$

where $\tilde{z}_t$ – the stationary component of productivity – evolves as in equation 2 while Z_t^p follows a non stationary process.¹ Specifically, $z_t^p = \log(Z_t^p/Z_{t-1}^p)$ follows AR(1) processes:

$$z_t^p = \rho_{zp} z_{t-1}^p + \sigma_{zp} \epsilon_{zp,t}, \quad \epsilon_{zp,t} \sim N(0, 1), \quad (23)$$

It follows that

$$z_t = \log(Z_t/Z_{t-1}) - \gamma = \frac{1}{1-\alpha} (\rho_z - 1) \tilde{z}_{t-1} + \frac{1}{1-\alpha} \sigma_z \epsilon_{z,t} + z_t^p, \quad (24)$$

and

$$E_t[z_{t+1}] = \frac{1}{1-\alpha} (\rho_z - 1) \tilde{z}_t + \rho_{zp} z_t^p. \quad (25)$$

Since there is a stochastic trend in growth, in equations 11 and 12 the term $\frac{1}{1-\alpha} \tilde{z}_t$ needs to be dropped.

¹The production function is $Y_t(i) = \max\{e^{\tilde{z}_t} K_t(i)^\alpha (L_t(i) e^{\gamma t} Z_t^p)^{1-\alpha} - \Phi Z_t, 0\}$.

1.5 Anticipated Policy Shocks

This section describes the introduction of anticipated policy shocks in the model, which follows Laseen and Svensson (2011). We modify the exogenous component of the policy rule (17) so to incorporate anticipated policy shocks:

$$r_t^m = \rho_{r^m} r_{t-1}^m + \epsilon_t^R + \sum_{k=1}^K \epsilon_{k,t-k}^R, \quad (26)$$

where $\epsilon_{R,t}$ is the usual contemporaneous policy shock, and $\epsilon_{k,t-k}^R$ is a policy shock that is known to agents at time $t-k$, but affects the policy rule k periods later, that is, at time t . We assume that $\epsilon_{k,t-k}^R \sim N(0, \sigma_{k,r}^2)$, *i.i.d.*

In order to solve the model we need to express the anticipated shocks in recursive form. For this purpose, we augment the state vector s_t with K additional states $\nu_t^R, \dots, \nu_{t-K}^R$ whose law of motion is as follows:

$$\begin{aligned} \nu_{1,t}^R &= \nu_{2,t-1}^R + \epsilon_{1,t}^R \\ \nu_{2,t}^R &= \nu_{3,t-1}^R + \epsilon_{2,t}^R \\ &\vdots \\ \nu_{K,t}^R &= \epsilon_{K,t}^R \end{aligned}$$

and rewrite the policy rule (26) as²

$$r_t^m = \rho_{r^m} r_{t-1}^m + \epsilon_t^R + \nu_{1,t-1}^R.$$

1.6 State Space Representation

We use the method in Sims (2002) to solve the system of log-linear approximate equilibrium conditions and obtain the transition equation:

$$s_t = \mathcal{T}(\theta) s_{t-1} + \mathcal{R}(\theta) \epsilon_t. \quad (27)$$

We collect all the DSGE model parameters in a vector θ and stack the structural shocks in a vector ϵ_t . The state-space representation for our vector of observables y_t , which we describe in the next section, is composed of the transition equation (27), which summarizes

²It is easy to verify that $\nu_{1,t-1}^R = \sum_{k=1}^K \epsilon_{k,t-k}^R$, that is, $\nu_{1,t-1}^R$ is a “bin” that collects all anticipated shocks that affect the policy rule in period t .

the evolution of the states s_t , and of a system of measurement equations:

$$y_t = \mathcal{D}(\theta) + \mathcal{Z}(\theta)s_t, \quad (28)$$

mapping the states into the observables, which we describe in detail in the section ???. We assume that some of the variables are measured with “error”, that is, the observed value equals the model implied value plus an exogenous process, which evolves as an AR(1). We add this exogenous process to the vector of states s_t .

3 Data

The estimation of the DSGE model is based on data on real output growth, consumption growth, investment growth, real wage growth, hours worked, inflation, interest rates, 10-year inflation expectations, and spreads. Measurement equations related the model variables that appeared in Section 2 to the observables:

$$\begin{aligned} \text{Output growth} &= \gamma + 100 (y_t - y_{t-1} + z_t) \\ \text{Consumption growth} &= \gamma + 100 (c_t - c_{t-1} + z_t) \\ \text{Investment growth} &= \gamma + 100 (i_t - i_{t-1} + z_t) \\ \text{Real Wage growth} &= \gamma + 100 (w_t - w_{t-1} + z_t) \\ \text{Hours} &= \bar{l} + 100l_t \\ \text{Core PCE Inflation} &= \pi_* + 100\pi_t + e_{pce,t} \\ \text{GDP Deflator Inflation} &= \pi_* + \lambda_{gdp} * 100\pi_t + e_{gdp,t} \\ \text{FFR} &= R_* + 100R_t \\ \text{10y Nominal Bond Yield} &= R_* + 100\mathbb{E}_t \left[\frac{1}{40} \sum_{k=1}^{40} R_{t+k} \right] + e_{R,t} \\ \text{10y Infl. Expectations} &= \pi_* + 100\mathbb{E}_t \left[\frac{1}{40} \sum_{k=1}^{40} \pi_{t+k} \right] \\ \text{Spread} &= SP_* + 100\mathbb{E}_t \left[\tilde{R}_{t+1}^k - R_t \right] \\ \text{TFP growth} &= \gamma + 100 (z_t + u_t) + e_{z,t} \end{aligned} \quad (29)$$

where all variables are measured in percent. All the $e_{*,t}$ processes follow an exogenous AR(1) specification, and can be thought of either measurement error or some other unmodeled source of discrepancy between the model and the data (e.g., risk premia for the long term nominal rate). The terms π_* and R_* measure the steady state level of net inflation and short term nominal interest rates, respectively, and $\bar{l}$ captures the mean of hours (this variable is

measured as an index). The first seven series are commonly used in the estimation of the SW model. The 10-year inflation expectations contain information about low-frequency inflation movements and are obtained from the Blue Chip Economic Indicators survey and the Survey of Professional Forecasters. As spread variable we use a Baa Corporate Bond Yield spread over the 10-Year Treasury Note Yield at constant maturity. Details on the construction of the data set are provided in Appendix A.

In order to estimate the importance of anticipated shocks and their effect on the variables of interest, we follow Del Negro and Schorfheide (2012) and augment the measurement equation (28) with the expectations for the policy rate:

$$\begin{aligned} FFR_{t,t+1}^e &= 100(\mathcal{Z}(\theta)_{R,\cdot} \mathcal{T}(\theta)^1 s_t + R^*), \\ &\vdots \\ FFR_{t,t+K}^e &= 100(\mathcal{Z}(\theta)_{R,\cdot} \mathcal{T}(\theta)^K s_t + R^*), \end{aligned} \tag{30}$$

where $FFR_{t,t+k}^e$ are the market's expectations for the FFR k quarters ahead, and $\mathcal{Z}(\theta)_{R,\cdot}$ is the row of $\mathcal{Z}(\theta)$ corresponding to the interest rate.

4 Inference, Prior and Posterior Parameter Estimates

We use Bayesian techniques in the subsequent empirical analysis, which require the specification of a prior distribution for the model parameters. For most of the parameters we use the same marginal prior distributions as Smets and Wouters (2007). There are two important exceptions. First, the original prior for the quarterly steady state inflation rate π_* used by Smets and Wouters (2007) is tightly centered around 0.62% (which is about 2.5% annualized) with a standard deviation of 0.1%. We favor a looser prior, one that has less influence on the model's forecasting performance, that is centered at 0.75% and has a standard deviation of 0.4%. Second, for the financial frictions mechanism we specify priors for the parameters SP_* , $\zeta_{sp,b}$, ρ_{σ_ω} , and σ_{σ_ω} . We fix the parameters corresponding to the steady state default probability and the survival rate of entrepreneurs, respectively. In turn, these parameters imply values for the parameters of (21).

Information on the the prior and posterior mean is provided in section B. Section C reports the impulse response functions of the observable variables to the various shocks.

References

- ARUOBA, S. B. AND F. SCHORFHEIDE (2008): “Insights from an Estimated Search-Based Monetary Model with Nominal Rigidities,” *Working Paper*.
- BERNANKE, B., M. GERTLER, AND S. GILCHRIST (1999a): “The Financial Accelerator in a Quantitative Business Cycle Framework,” in *Handbook of Macroeconomics*, ed. by J. B. Taylor and M. Woodford, North Holland, Amsterdam, vol. 1C.
- BERNANKE, B. S., M. GERTLER, AND S. GILCHRIST (1999b): “The Financial Accelerator in a Quantitative Business Cycle Framework,” in *Handbook of Macroeconomics*, ed. by J. B. Taylor and M. Woodford, Amsterdam: North-Holland, vol. 1C, chap. 21, 1341–93.
- CHRISTIANO, L., R. MOTTO, AND M. ROSTAGNO (2003): “The Great Depression and the Friedman-Schwartz Hypothesis,” *Journal of Money, Credit and Banking*, 35, 1119–1197.
- CHRISTIANO, L. J., M. EICHENBAUM, AND C. L. EVANS (2005): “Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy,” *Journal of Political Economy*, 113, 1–45.
- CHRISTIANO, L. J., R. MOTTO, AND M. ROSTAGNO (2014): “Risk Shocks,” *American Economic Review*, 104, 27–65.
- DE GRAEVE, F. (2008): “The External Finance Premium and the Macroeconomy: US Post-WWII Evidence,” *Journal of Economic Dynamics and Control*, 32, 3415 – 3440.
- DEL NEGRO, M. AND S. EUSEPI (2011): “Fitting Observed Inflation Expectations,” *Journal of Economic Dynamics and Control*, 35, 2105–2131.
- DEL NEGRO, M., M. P. GIANNONI, AND F. SCHORFHEIDE (2015): “Inflation in the Great Recession and New Keynesian Models,” *American Economic Journal: Macroeconomics*, 7, 168–196.
- DEL NEGRO, M. AND F. SCHORFHEIDE (2012): “DSGE Model-Based Forecasting,” *FRBNY Working Paper*.
- ERCEG, C. J. AND A. T. LEVIN (2003): “Imperfect Credibility and Inflation Persistence,” *Journal of Monetary Economics*, 50, 915–944.

- GREENWOOD, J., Z. HERCOVITZ, AND P. KRUSELL (1998): “Long-Run Implications of Investment-Specific Technological Change,” *American Economic Review*, 87, 342–36.
- LASEEN, S. AND L. E. SVENSSON (2011): “Anticipated Alternative Policy-Rate Paths in Policy Simulations,” *International Journal of Central Banking*, 7, 1–35.
- PRIMICERI, G. (2006): “Why Inflation Rose and Fell: Policymakers Beliefs and US Postwar Stabilization Policy,” *Quarterly Journal of Economics*, 121, 867–901.
- SARGENT, T. J. (1999): *The Conquest of American Inflation*, Princeton University Press, Princeton.
- SIMS, C. A. (2002): “Comment on Cogley and Sargent’s ‘Evolving post World War II U.S. Inflation Dynamics’ ,” in *NBER Macroeconomics Annual 2001*, ed. by B. S. Bernanke and K. Rogoff, MIT Press, Cambridge, vol. 16, 373–379.
- SMETS, F. AND R. WOUTERS (2003): “An Estimated Dynamic Stochastic General Equilibrium Model of the Euro Area,” *Journal of the European Economic Association*, 1, 1123 – 1175.
- (2007): “Shocks and Frictions in US Business Cycles: A Bayesian DSGE Approach,” *American Economic Review*, 97, 586 – 606.

A Data Construction

Data on real GDP (*GDPC*), the GDP deflator (*GDPDEF*), core PCE inflation (*PCEPILFE*), nominal personal consumption expenditures (*PCEC*), and nominal fixed private investment (*FPI*) are produced at a quarterly frequency by the Bureau of Economic Analysis, and are included in the National Income and Product Accounts (NIPA). Average weekly hours of production and nonsupervisory employees for total private industries (*AWHNONAG*), civilian employment (*CE16OV*), and the civilian non-institutional population (*LNSINDEX*) are produced by the Bureau of Labor Statistics (BLS) at a monthly frequency. The first of these series is obtained from the Establishment Survey, and the remaining from the Household Survey. Both surveys are released in the BLS Employment Situation Summary. Since our models are estimated on quarterly data, we take averages of the monthly data. Compensation per hour for the non-farm business sector (*COMPNFB*) is obtained from the Labor Productivity and Costs release, and produced by the BLS at a quarterly frequency. All data are transformed following Smets and Wouters (2007). The federal funds rate is obtained from the Federal Reserve Board's H.15 release at a business day frequency. We take quarterly averages of the annualized daily data and divide by four. Let Δ denote the temporal difference operator. Then:

$$\begin{aligned}
 \text{Output growth} &= 100 * \Delta \ln((GDPC)/LNSINDEX) \\
 \text{Consumption growth} &= 100 * \Delta \ln((PCEC/GDPDEF)/LNSINDEX) \\
 \text{Investment growth} &= 100 * \Delta \ln((FPI/GDPDEF)/LNSINDEX) \\
 \text{Real wage growth} &= 100 * \Delta \ln(COMP NFB/GDPDEF) \\
 \text{Hours worked} &= 100 * \Delta \ln((AWHNONAG * CE16OV/100)/LNSINDEX) \\
 \text{GDP Deflator Inflation} &= 100 * \Delta \ln(GDPDEF) \\
 \text{Core PCE Inflation} &= 100 * \Delta \ln(PCEPILFE) \\
 \text{FFR} &= (1/4) * FEDERAL FUNDS RATE
 \end{aligned}$$

Long-run inflation expectations are obtained from the Blue Chip Economic Indicators survey and the Survey of Professional Forecasters available from the FRB Philadelphia's Real-Time Data Research Center. Long-run inflation expectations (average CPI inflation over the next 10 years) are available from 1991Q4 onward. Prior to 1991Q4, we use the 10-year expectations data from the Blue Chip survey to construct a long time series that begins in 1979Q4. Since the Blue Chip survey reports long-run inflation expectations only

twice a year, we treat these expectations in the remaining quarters as missing observations and adjust the measurement equation of the Kalman filter accordingly. Long-run inflation expectations $\pi_t^{O,40}$ are therefore measured as

$$10y \text{ Infl Exp} = (10\text{-YEAR AVERAGE CPI INFLATION FORECAST} - 0.50)/4.$$

where 0.50 is the average difference between CPI and GDP annualized inflation from the beginning of the sample to 1992. We divide by 4 to express the data in quarterly terms. Finally, we measure *Spread* as the annualized Moody's Seasoned Baa Corporate Bond Yield spread over the 10-Year Treasury Note Yield at Constant Maturity. Both series are available from the Federal Reserve Board's H.15 release. Like the federal funds rate, the spread data are also averaged over each quarter and measured at a quarterly frequency. This leads to:

$$\text{Spread} = (1/4) * (\text{Baa Corporate} - 10 \text{ year Treasury}).$$

Similarly,

$$10y \text{ Bond yield} = (1/4) * (10 \text{ year Treasury}).$$

Last TFP growth is measured using John Fernald's TFP growth, unadjusted for changes in utilization and expressed in labor augmenting terms:

$$\text{TFP growth} = (1/4) * \text{Fernald's TFP growth, unadjusted}.$$

B Prior and Posterior Distributions

Parameter Estimates: Prior and Posterior Mean

Parameter	Prior Mean	Prior Stdd	Post Mean	90% Lower Band	90% Upper Band
α	0.300	0.050	0.167	0.143	0.191
ι_p	0.500	0.150	0.256	0.115	0.398
Υ	1.000	0.000	1.000	1.000	1.000
Φ	1.250	0.120	1.120	1.049	1.189
S''	4.000	1.500	2.940	1.941	3.874
h	0.700	0.100	0.440	0.353	0.530
ψ	0.500	0.150	0.725	0.586	0.866
ι_w	0.500	0.150	0.526	0.314	0.738
β	0.250	0.100	0.132	0.055	0.202
σ_c	1.500	0.370	1.120	0.878	1.353
ρ	0.750	0.100	0.665	0.599	0.733
$F(\omega)$	0.030	0.000	0.030	0.030	0.030
spr_*	2.000	0.100	1.752	1.625	1.879
γ_*	0.990	0.000	0.990	0.990	0.990
γ	0.400	0.100	0.348	0.277	0.423
$Lmean$	-45.000	5.000	-47.469	-49.289	-45.633
σ_g	0.100	2.000	2.559	2.352	2.760
σ_b	0.100	2.000	0.029	0.025	0.034
σ_μ	0.100	2.000	0.468	0.403	0.533
σ_z	0.100	2.000	0.678	0.612	0.743
σ_{λ_f}	0.100	2.000	0.085	0.066	0.102
σ_{λ_w}	0.100	2.000	0.384	0.337	0.431
σ_{rm}	0.100	2.000	0.235	0.213	0.258
σ_{sigw}	0.050	4.000	0.043	0.034	0.051

Parameter Estimates: Prior and Posterior Mean

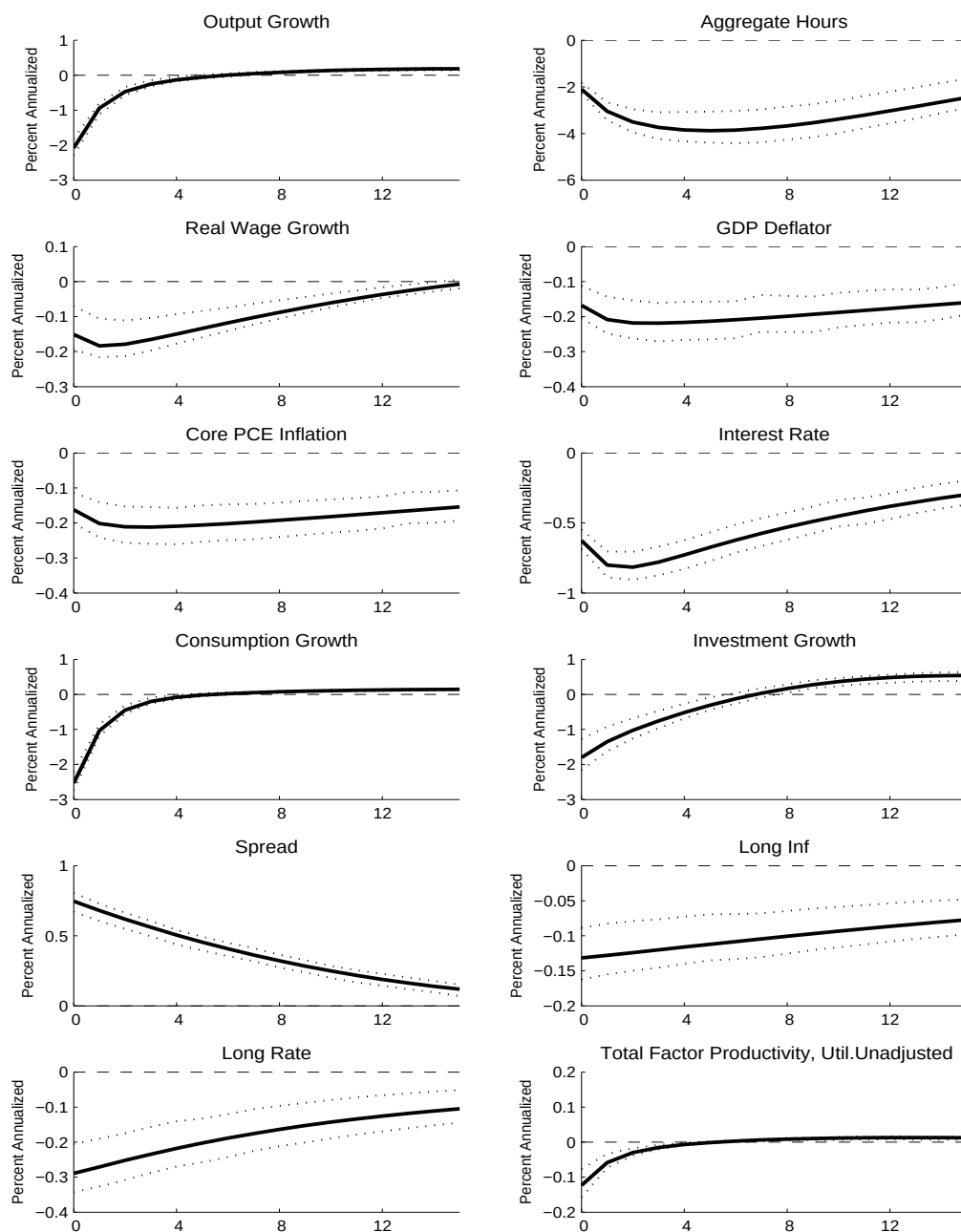
Parameter	Prior Mean	Prior Stdd	Post Mean	90% Lower Band	90% Upper Band
σ_{mue}	0.000	0.000	0.000	0.000	0.000
σ_{gamm}	0.000	0.000	0.000	0.000	0.000
σ_{lr}	0.750	2.000	0.184	0.149	0.219
σ_{zp}	0.100	2.000	0.194	0.104	0.288
σ_{tfp}	0.100	2.000	0.828	0.755	0.899
σ_{gdpdef}	0.100	2.000	0.160	0.143	0.179
σ_{pce}	0.100	2.000	0.100	0.081	0.120
σ_{ant1}	0.200	4.000	0.100	0.076	0.122
σ_{ant2}	0.200	4.000	0.090	0.070	0.110
σ_{ant3}	0.200	4.000	0.090	0.069	0.110
σ_{ant4}	0.200	4.000	0.085	0.065	0.103
σ_{ant5}	0.200	4.000	0.086	0.066	0.105
σ_{ant6}	0.200	4.000	0.109	0.083	0.134
σ_{ant7}	0.200	4.000	0.258	0.105	0.400
σ_{ant8}	0.200	4.000	0.309	0.104	0.474
σ_{ant9}	0.200	4.000	0.251	0.106	0.399
σ_{ant10}	0.200	4.000	0.250	0.101	0.399
σ_{ant11}	0.200	4.000	0.234	0.112	0.362
σ_{ant12}	0.200	4.000	0.236	0.108	0.366
σ_{ant13}	0.000	0.000	0.000	0.000	0.000
σ_{ant14}	0.000	0.000	0.000	0.000	0.000
σ_{ant15}	0.000	0.000	0.000	0.000	0.000
σ_{ant16}	0.000	0.000	0.000	0.000	0.000
σ_{ant17}	0.000	0.000	0.000	0.000	0.000
σ_{ant18}	0.000	0.000	0.000	0.000	0.000

Parameter Estimates: Prior and Posterior Mean

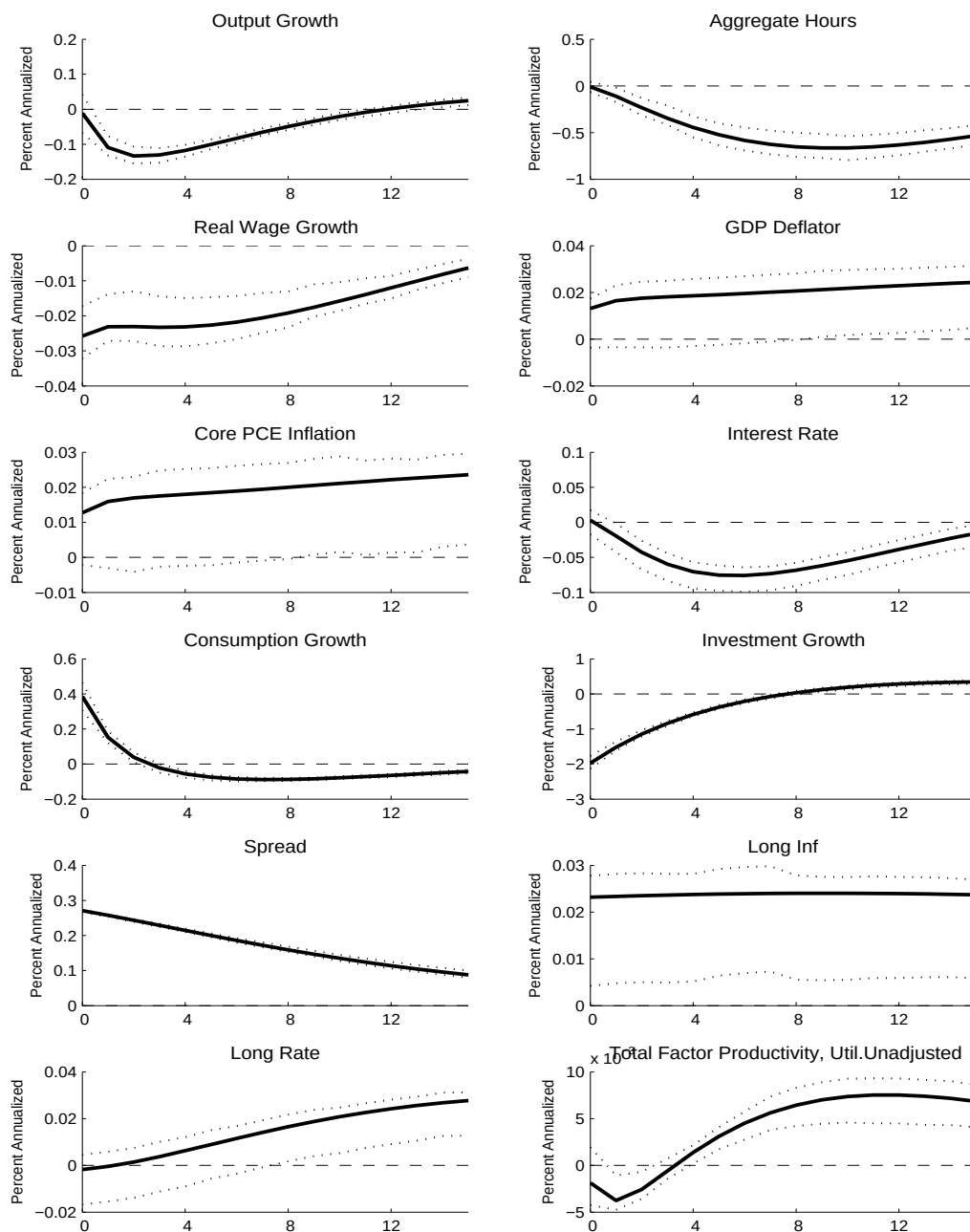
Parameter	Prior Mean	Prior Stdd	Post Mean	90% Lower Band	90% Upper Band
σ_{ant19}	0.000	0.000	0.000	0.000	0.000
σ_{ant20}	0.000	0.000	0.000	0.000	0.000
η_{gz}	0.500	0.200	0.768	0.580	0.962
η_{λ_f}	0.500	0.200	0.608	0.401	0.822
η_{λ_w}	0.500	0.200	0.432	0.184	0.683
i_{α}^{model}	0.000	0.000	0.000	0.000	0.000
Γ_{gdpdef}	1.000	2.000	1.033	0.951	1.118
δ_{gdpdef}	0.000	2.000	0.000	-0.053	0.054

C Impulse Response Functions

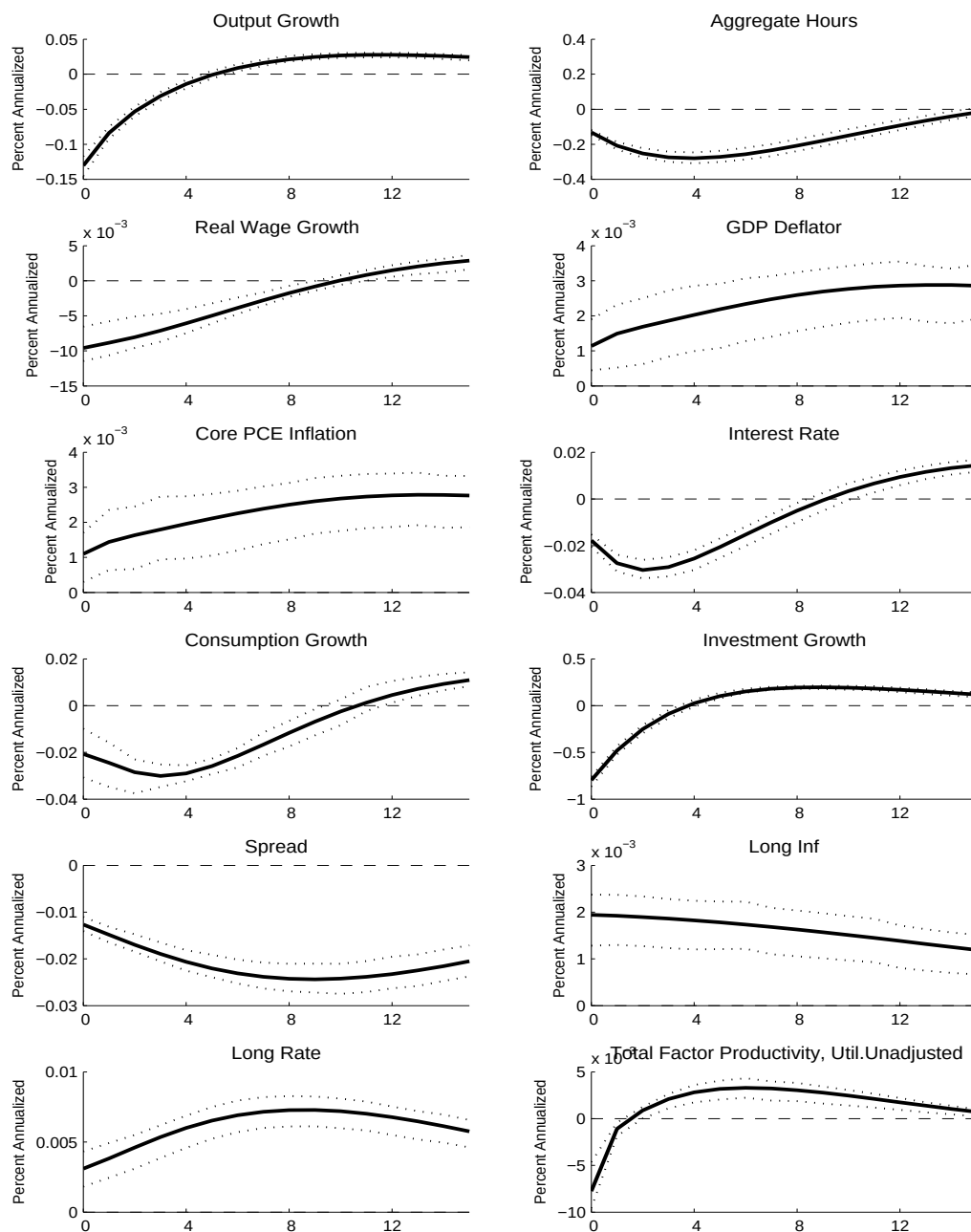
Responses to a Discount Rate (*b*) Shock



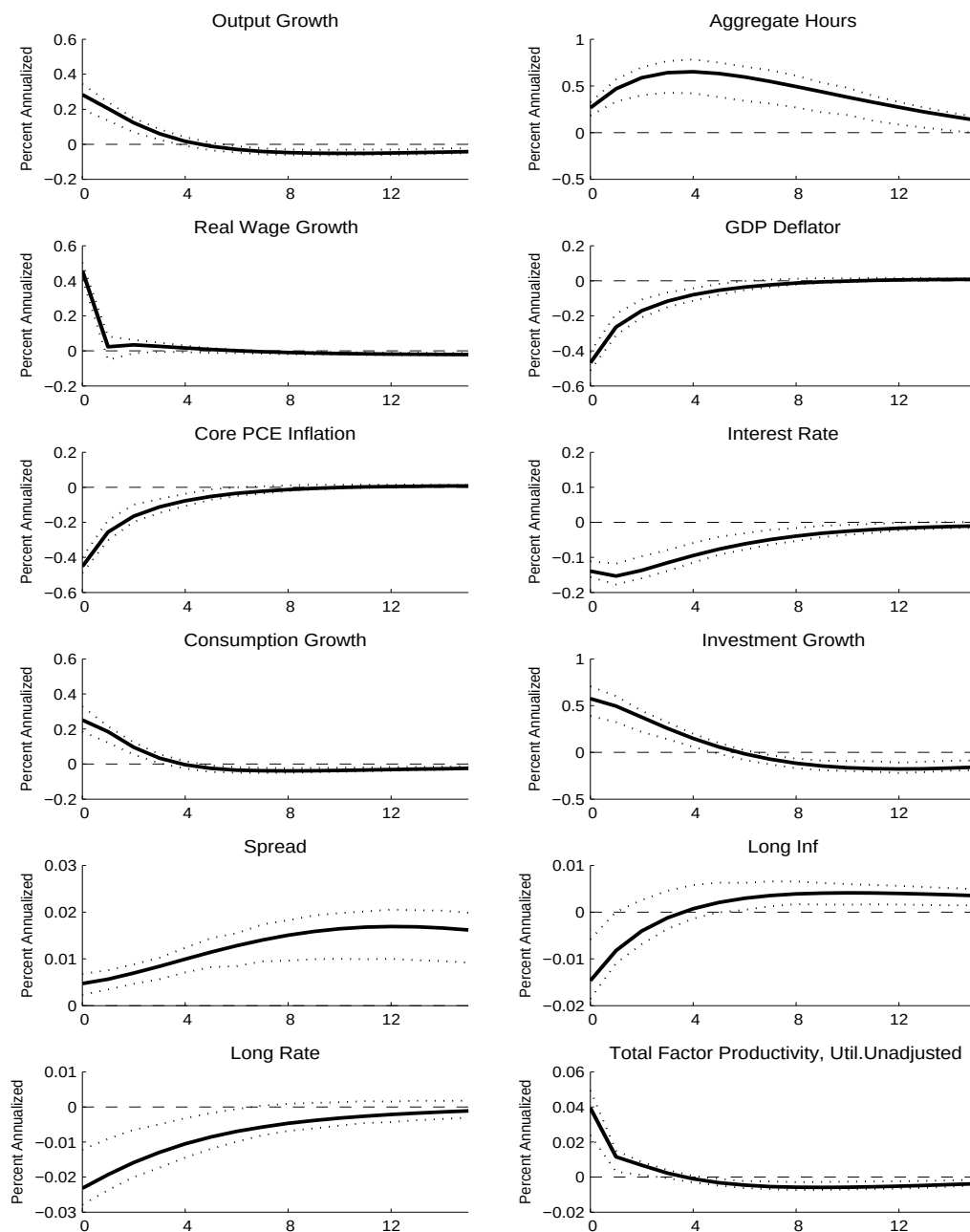
Responses to a Spread Shock



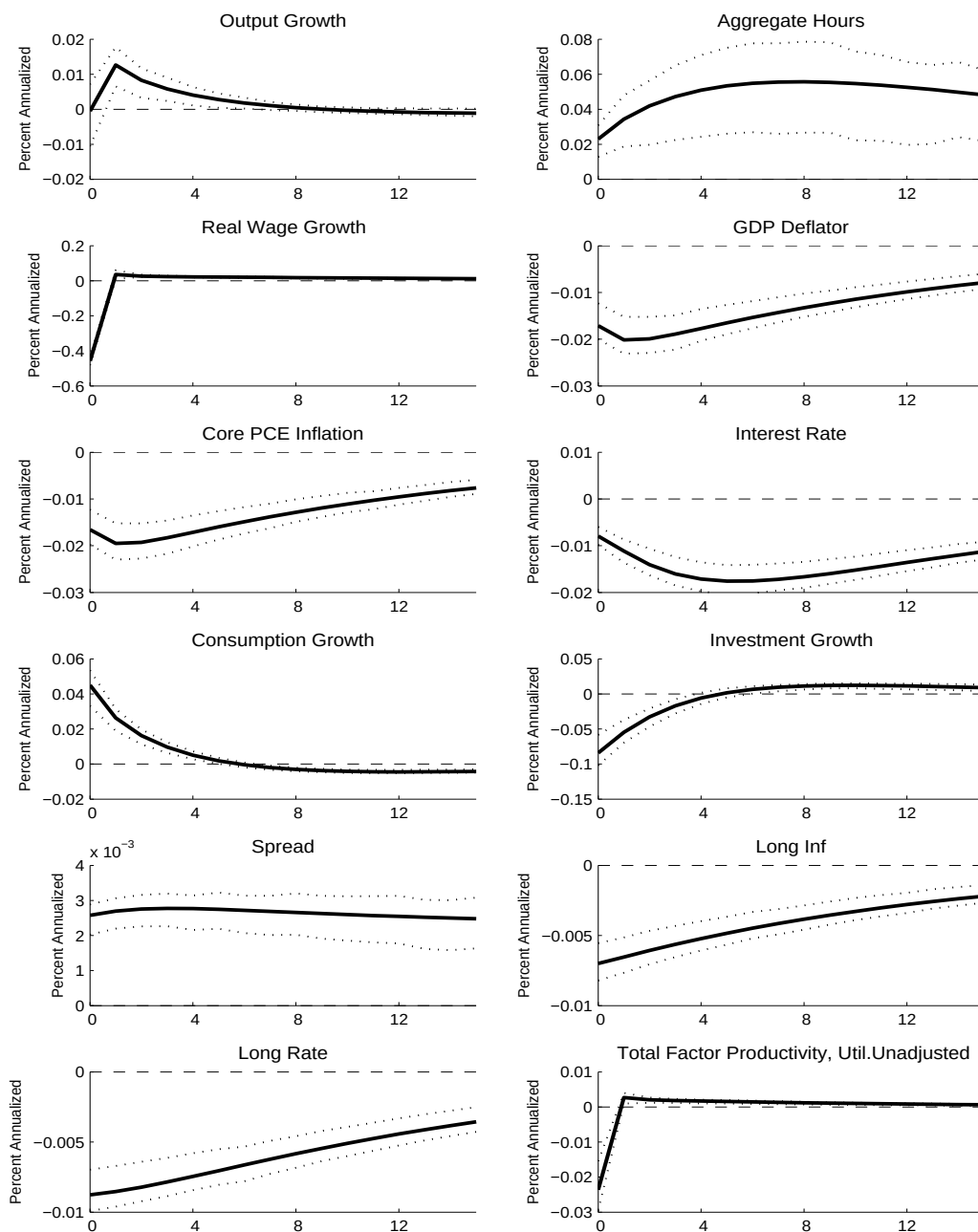
Responses to an MEI Shock



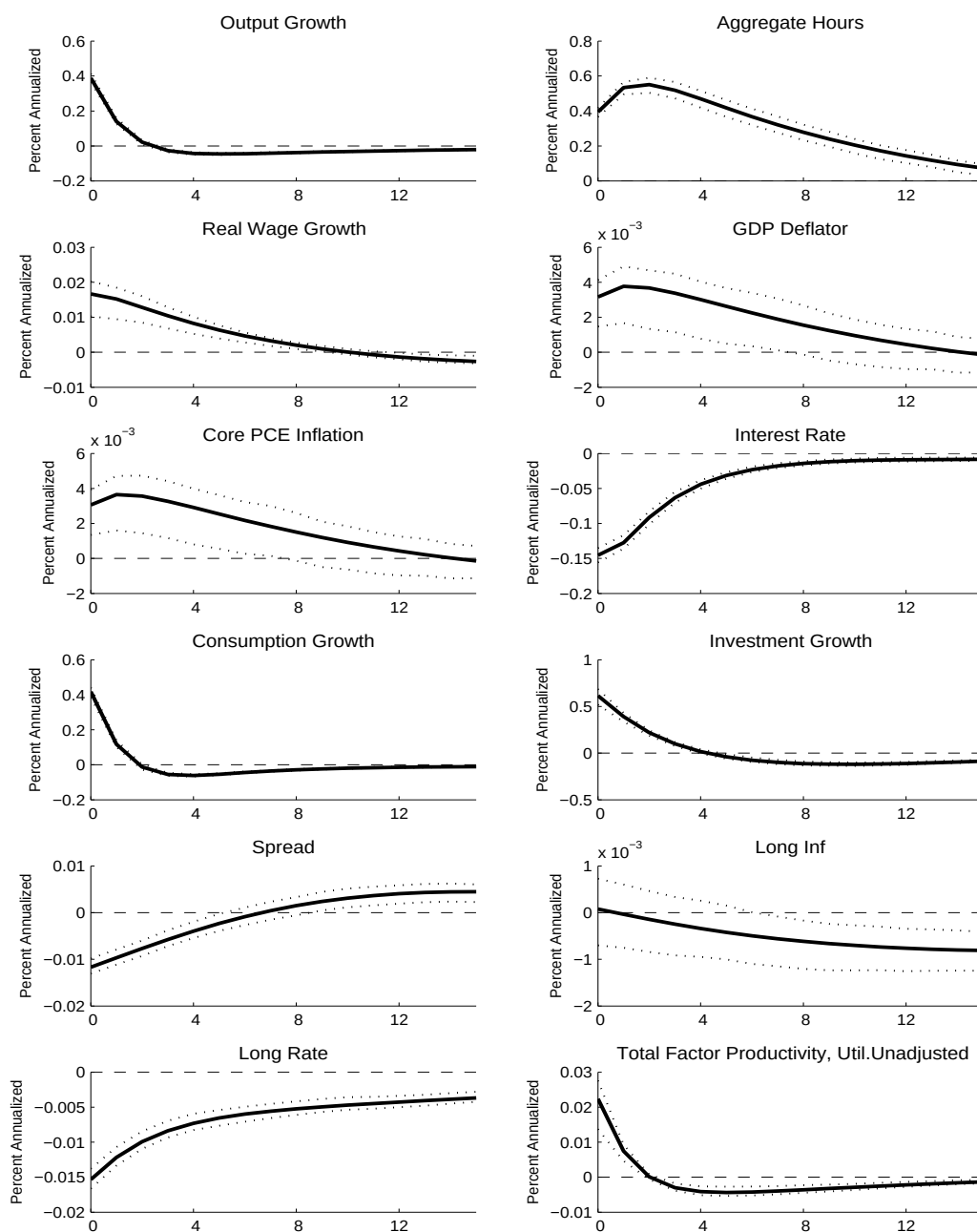
Responses to a Price Mark-up Shock



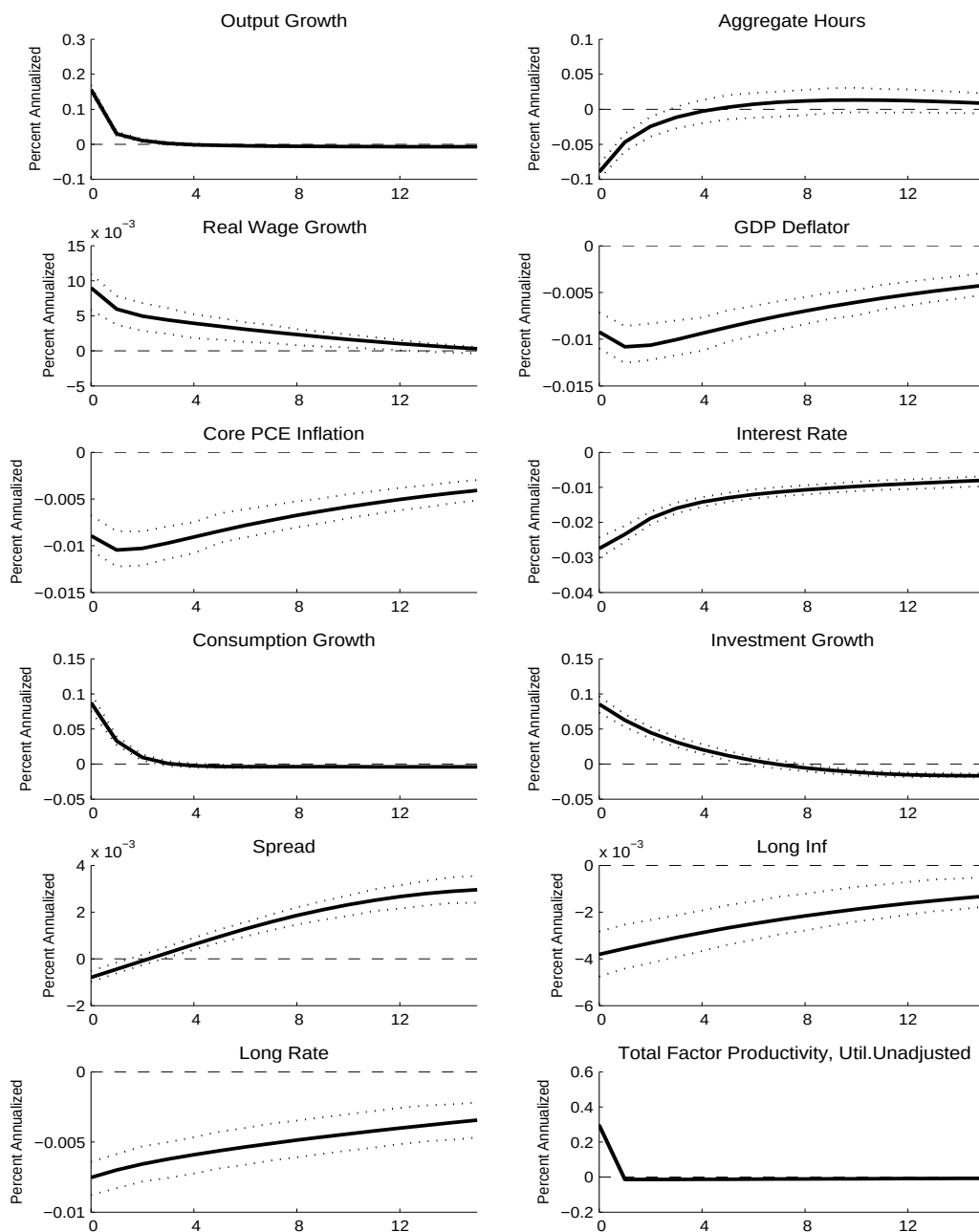
Responses to a Wage Mark-up Shock



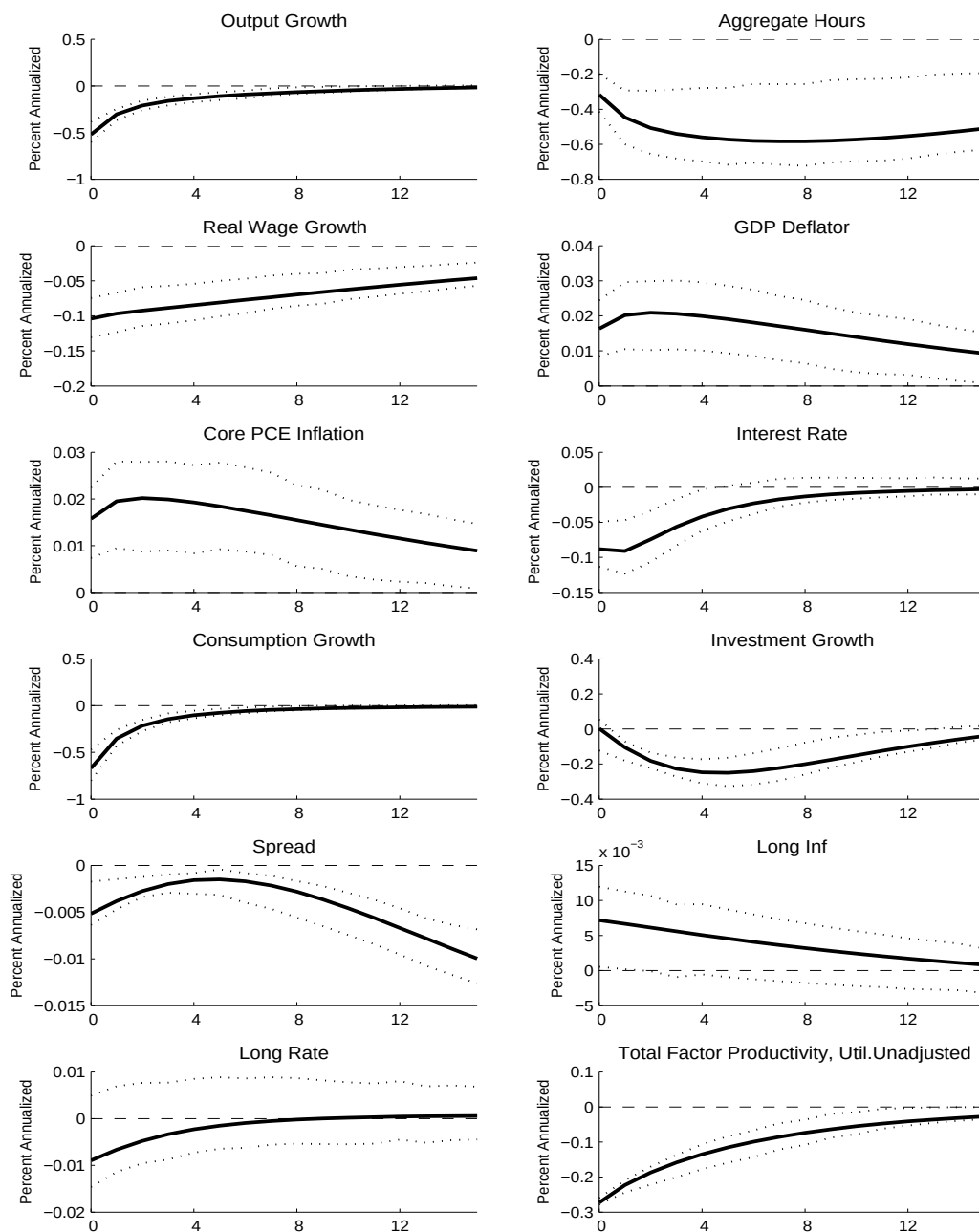
Responses to a Monetary Policy Shock



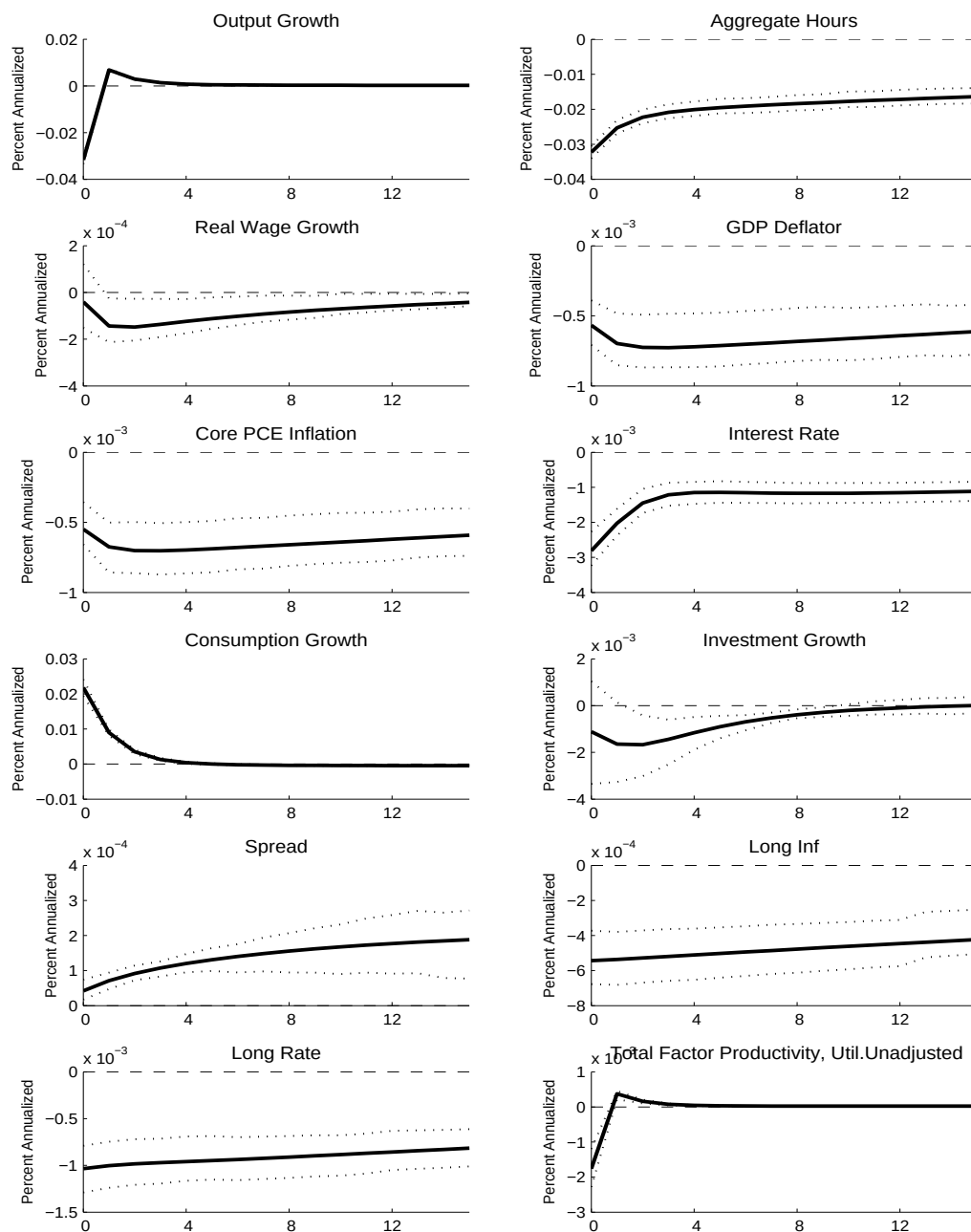
Responses to a Stationary Productivity Shock

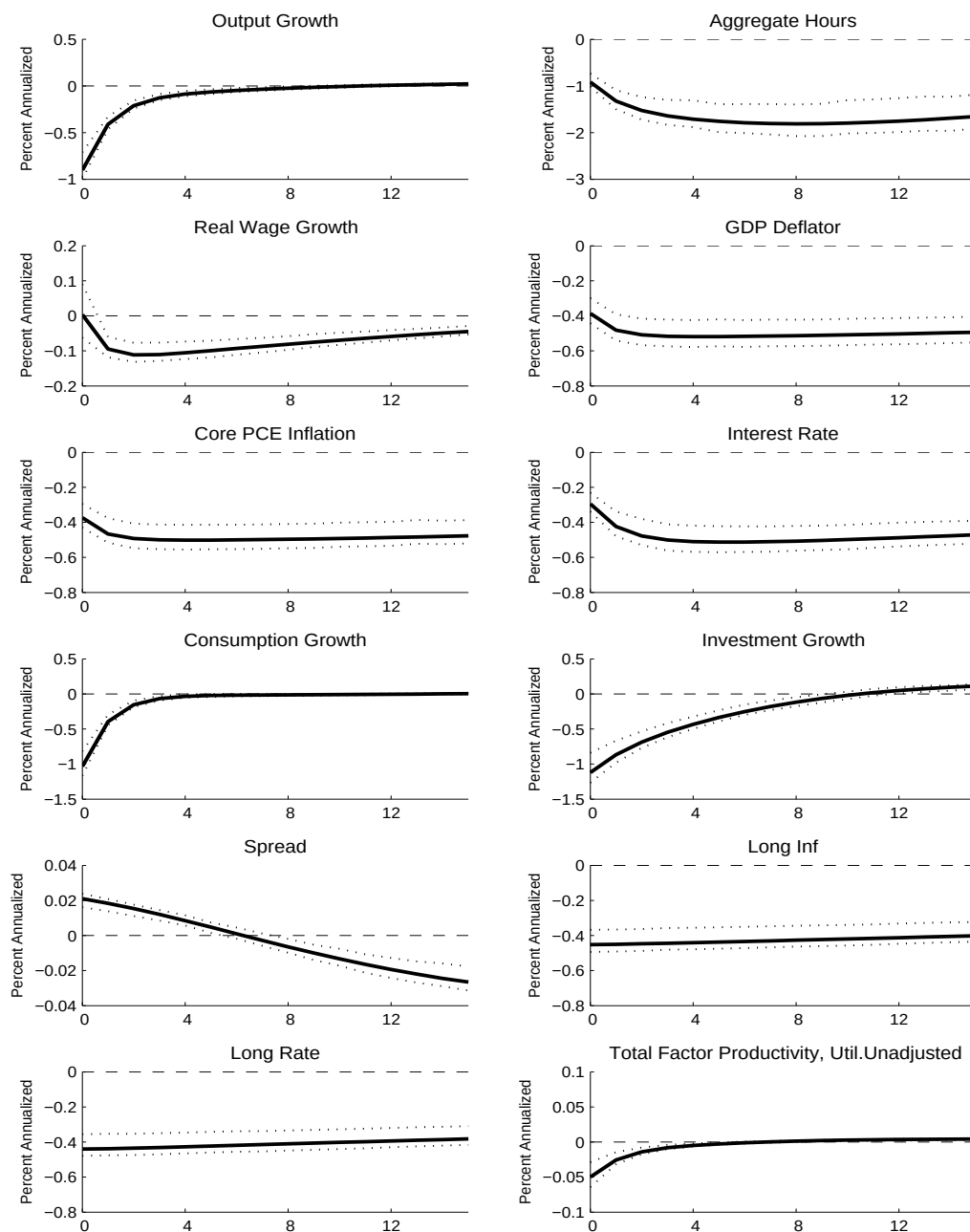


Responses to a Shock to the TFP Growth Rate

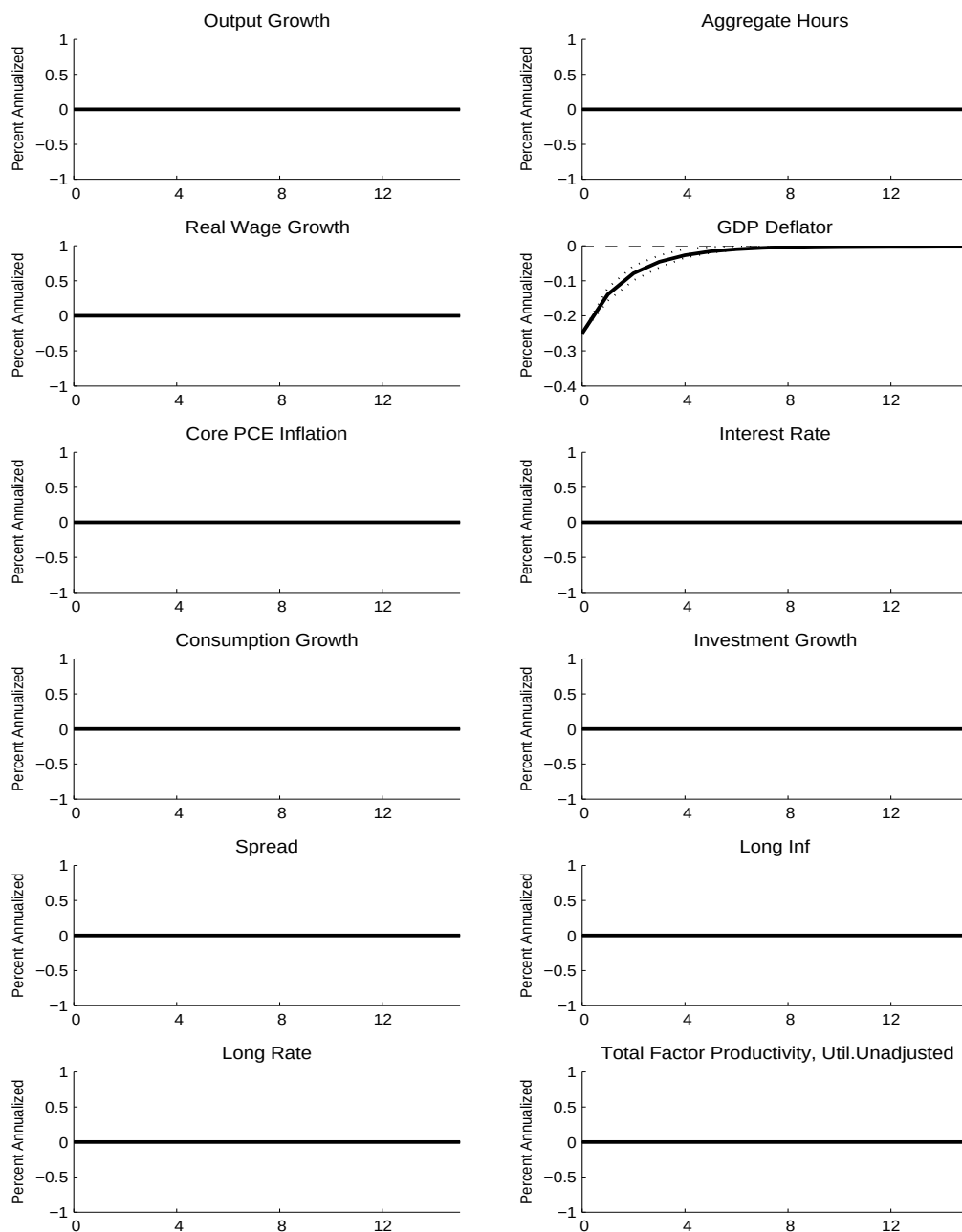


Responses to a Government Spending Shock

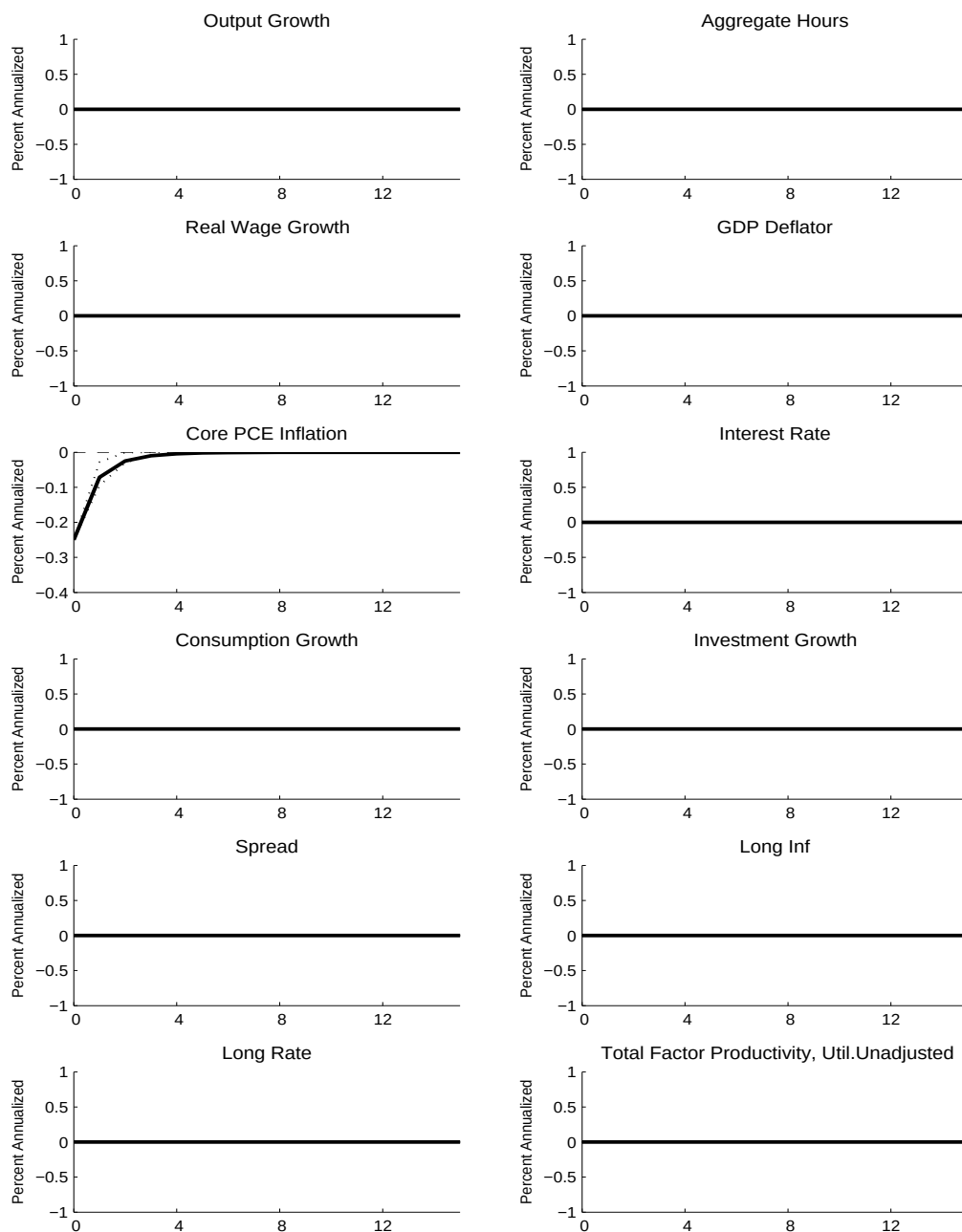


Responses to a π^* Shock

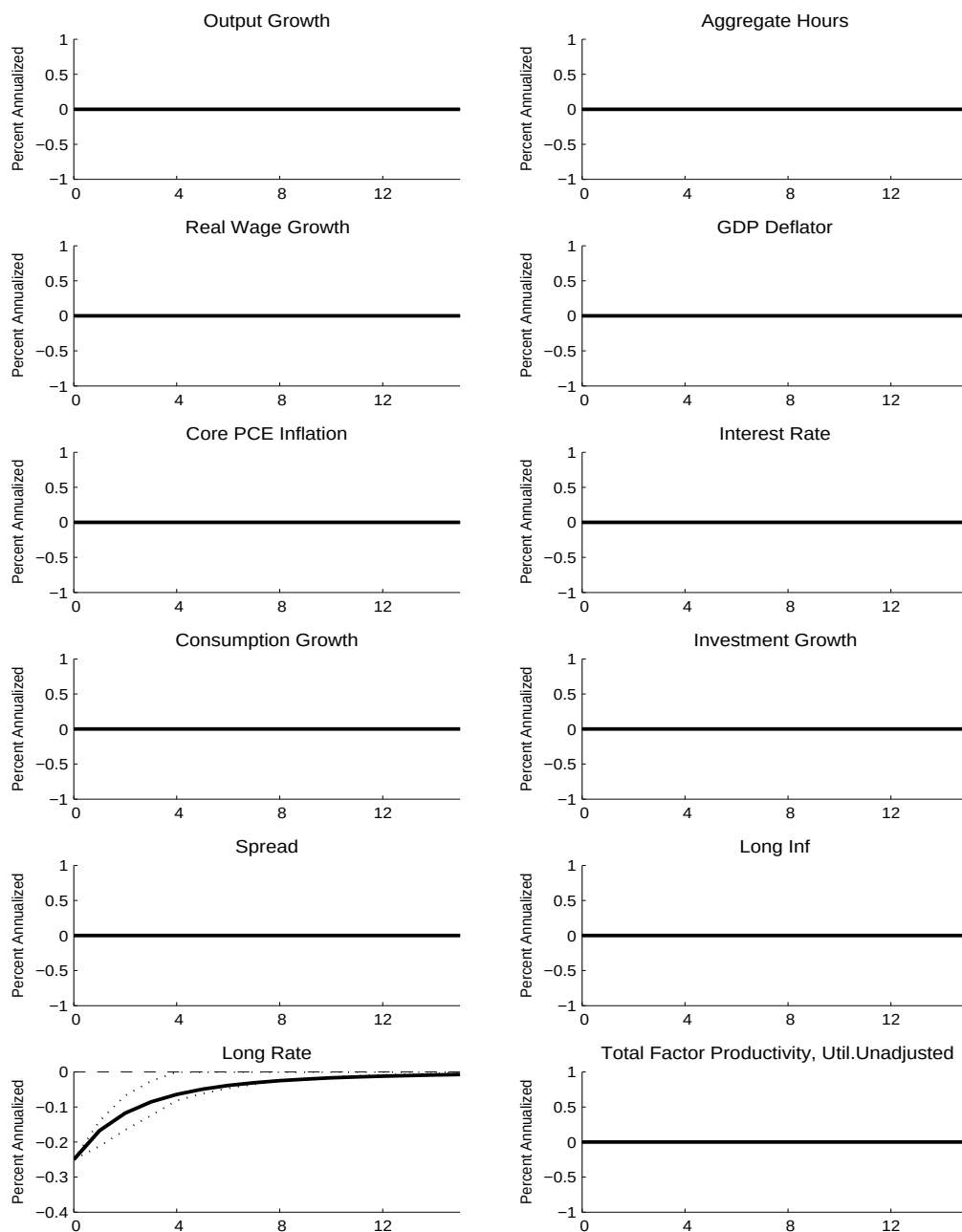
Responses to GDP Deflator Measurement Error



Responses to Core PCE Measurement Error



Responses to Long Rate Measurement Error



Responses to TFP Measurement Error

